



Preparation and electrical properties of 4-acetoxystyrene grafted polypropylene for HVDC cable insulation

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Abstract

Polypropylene (PP) is considered to be a potential eco-friendly cable insulation material. However, space charge accumulation in PP under high electric stress limits its application for HVDC cable. In this paper, PP was modified with 4-acetoxystyrene (AOS) by melt grafting to improve electric properties. The molecular structure of the polypropylene-graft-4-acetoxystyrene (PP-g-AOS) was characterized by means of Fourier Transform Infrared Spectroscopy (FT-IR). The effects of temperature, screw speed, time, initiator content and monomer content on grafting degree of PP-g-AOS were investigated. Furthermore, the influence of grafting degree of PP-g-AOS on space charge suppression, volume resistivity and dc breakdown field strength was investigated. The FT-IR result showed that AOS was successfully grafted onto PP. It was found that under the proper grafting conditions, the obtained PP-g-AOS exhibited a maximum grafting degree of 1.31%. Moreover, PP-g-AOS with grafting degree of 1.14% can suppress space charge efficiently, and increase both volume resistivity and dc breakdown strength. This provides a new idea for the study of polypropylene HVDC cable insulation.

1 Introduction

It is widely recognized that High voltage direct current (HVDC) transmission possesses the advantages of long-distance transmission, bulk-power delivery, asynchronous interconnections, processing costs and long submarine cable crossings [1–4]. Traditional HVDC cables employ extruded cross-linked polyethylene (XLPE) as the insulation material due to its favorable combination of low material costs, appropriate electrical and mechanical properties. However, XLPE is difficult to be recycled due to its thermosetting characteristic, and the peroxide-based cross-linking agents used in the manufacture of XLPE also cause environmental pollution [5, 6]. With the increasing awareness of environmental protection around the world, there is a need for

novel materials to replace XLPE for the next generation of extruded HVDC cables.

Polypropylene (PP) has attracted increasing attention of researchers due to its desirable characteristics. PP is a thermoplastic material and holds a higher melting point than polyethylene allowing it to withstand high operating temperatures without cross-linking. It is a promising alternative to traditional XLPE for cable applications [7–13]. Nevertheless, it is easy to accumulate space charge in the PP under HVDC, which can cause distortion of local electric field and then breakdown and even insulation failure finally [14–18]. The space charge accumulation hampers application of PP in HVDC cable. Some work has been done for improving space charge suppression performance of PP. The existing studies have revealed that addition of inorganic nano-oxide, such as TiO₂, MMT, ZnO into PP can effectively improve the space charge distribution characteristics [19–23]. Nonetheless, nanoparticles are liable to agglomeration due to the high active surface, which may bring about performance degradation of the insulation [24].

Molecular structure design, unlike nano-doping, may avoid the problem of poor dispersion and offer a new route to improve the electrical properties of material. Chen et al. proposed the asymmetric molecular chain structure in cross-linked network of epoxy resin by attaching ring-like alicyclic amine and linear polyether amine on phenol via Mannich

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reaction and obtained polymer film with high breakdown strength [25]. Zha et al. prepared polypropylene grafted with maleic anhydride by reactive extrusion. The results indicated that due to the presence of polar groups in MAH, the PP-g-MAH exhibited suppressed space charge accumulation and enhanced breakdown strength [26]. Grafting technology is indeed an effective way to improve space charge inhibition property of polypropylene, but there are few reports about this, which needs to be further developed.

In this paper, PP was modified by grafting monomer 4-acetoxystyrene (AOS). The molecular structure of the PP grafted with AOS (PP-g-AOS) was characterized by means of FT-IR. The effects of grafting conditions such as temperature, screw speed, reaction time, and amounts of initiator and monomer on the degree of grafting were studied. Furthermore, the influences of grafting degree of PP-g-AOS on electrical properties such as space charge distribution, dc volume resistivity and dc breakdown field strength were determined. It is of great value for the preparation of eco-friendly HVDC insulating material.

2 Experimental

2.1 Materials

Isotactic polypropylene (iPP) used in this study is commercial grade polypropylene (T30S) supplied by Sinopec Daqing Petrochemical Co., China. It was used in pellet form and dried before use. 4-acetoxystyrene (AOS) was purchased from Shanghai Aladdin biochemical technology Co., LTD., China. Xylene, acetone and dicumyl peroxide (DCP) were provided by Tianjin Fuyu Fine Chemical Co. LTD. All the chemicals are reagent grade and used as received.

2.2 Samples preparation of PP-g-AOS

Grafting of AOS monomer onto PP was carried out by reactive blending process using a HAPRO RM-200C torque rheometer. First, 50 g PP and certain amounts of AOS and DCP were weighed and premixed in the stirred tank. Second, these components were fed into the torque rheometer. The grafting reaction was performed under different process conditions. Figure 1 shows the grafting reaction mechanism of PP-g-AOS.

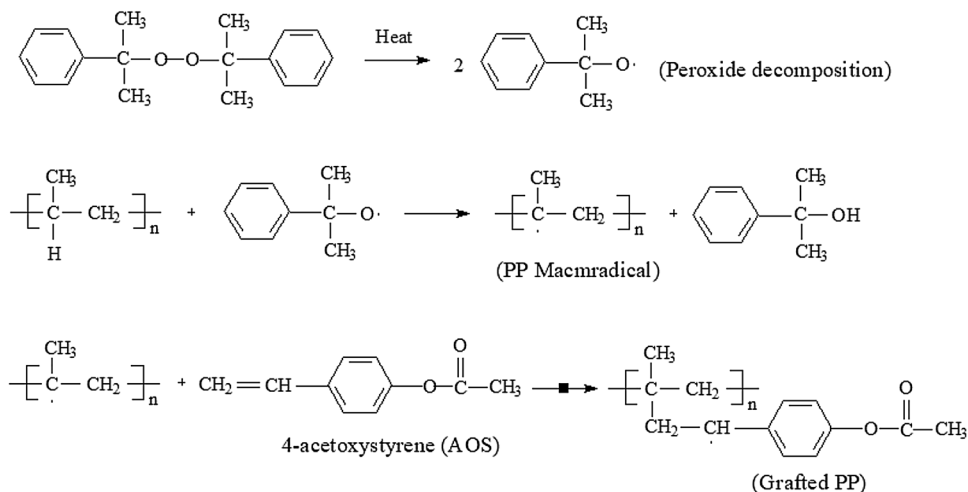
In order to remove the unreacted AOS and impurities, the grafted product was dissolved in refluxing xylene at 120 °C for 3 h and washed in acetone. Then, the PP-g-AOS was obtained. The resulting samples were hot-pressed under 190 °C into different thickness sheets using a plate vulcanizing machine for subsequent performance testing.

2.3 Characterization

Chemical structures of the pure PP and grafted PP were analyzed by Fourier Transform Infrared Spectroscopy (FT-IR) (Nicolet iS10, USA). Each sample was scanned 16 times with wave-number range from 4000 to 400 cm^{-1} at a resolution of 2 cm^{-1} . The peak at 2720 cm^{-1} corresponding to the PP backbone was chosen as the internal reference and the peak at 1762 cm^{-1} corresponding to the carbonyl group of AOS was used to determine grafting degree. The FT-IR absorption ratio of 1762–2720 cm^{-1} (I_{1762}/I_{2720}) was calculated, then grafting degree was obtained from a calibration curve established by using PP samples containing a known amount of AOS.

A FEI Sirion scanning electron microscope (SEM) was used to examine the surface morphologies of the unmodified PP and the grafted PP. SEM analysis was carried out using an acceleration voltage of 20 kV. The samples were

Fig. 1 PP-g-AOS grafting reaction mechanism



cryogenically fractured in liquid nitrogen and then coated with thin layers of gold prior to SEM examination.

X-ray diffraction (XRD) analysis was carried out on an Empyrean X-ray diffractometer with Cu K α radiation with a continuous scanning angle from 5° to 60°.

The space charge measurement was accomplished using a pulsed electro-acoustic (PEA) system equipped with a Lecroy WaveRunner 610Zi (1 GHz, 20 GS/s digital oscilloscope). The detailed steps were as follows. Before the measurement, aluminum was sputtered on both sides of the plate sample by vacuum evaporating. Then, the sample was short-circuited and was heat-treated in a vacuum at 80 °C for 24 h to eliminate the electric charges generated in the sample preparation process. After that, the sample was placed between the semi-conducting electrode (top electrode) and the aluminum electrode (bottom electrode), and it was required that the silicone oil was coated between the sample and both the electrodes. Finally, a dc electric field of 40 kV mm⁻¹ was applied to the sample for 30 min and the information about the space charge profiles under different duration was recorded.

The dc volume resistivity of PP-g-AOS sample was measured by a meter (ZC-36) with three-electrode system, according to ASTM D257-14.

DC breakdown strength was evaluated by a dielectric strength tester (CS2674). The specimen with thickness of ~100 μ m was sandwiched between two opposing plate electrodes. The lower electrode was earthed, and increased voltage at a boost rate of 1 kV/s was applied to the upper electrode until breakdown. During the experiment both the sample and the electrodes were immersed in silicone oil.

The breakdown field strength is calculated according to the Eq. (1):

$$E_B = \frac{U_B}{d} \quad (1)$$

where E_B is breakdown field strength (MV/m), U_B is breakdown voltage (MV), d is sample thickness at breakdown spot (m).

For each sample, 15 points were tested. The data analysis was conducted by two-parameter Weibull statistical distribution.

3 Results and discussion

3.1 FT-IR analysis

The existence of grafting AOS onto PP was confirmed by FT-IR spectroscopy. Figure 2 displays the FT-IR spectra of unmodified PP and PP-g-AOS. As shown in Fig. 2, the absorption bands at 1455 cm⁻¹ and 1375 cm⁻¹ are attributed

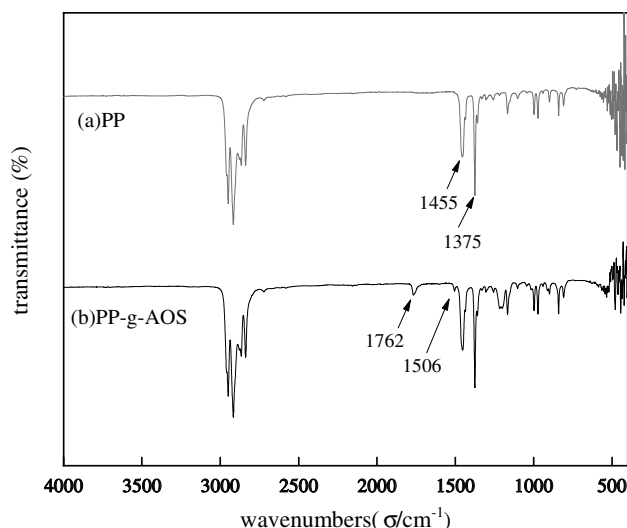


Fig. 2 FT-IR spectra of PP and PP-g-AOS

to C–H bending vibration of PP. On the PP-g-AOS spectral curve, it appears two additional absorption bands at 1762 cm⁻¹ assigned to stretching vibrations of the carbonyl group and at 1506 cm⁻¹ assigned to aromatic carbon–carbon (C=C) skeletal vibration of the monomer. Moreover, no absorption band near 1600 cm⁻¹ corresponding to vinyl carbon–carbon double bond stretching vibration was found near 1600 cm⁻¹. It is indicative of grafting AOS onto PP successfully.

3.2 SEM analysis

Figure 3 displays the surface morphologies of unmodified PP and PP-g-AOS. The surface of unmodified PP is very smooth as shown in Fig. 3a. However, after the graft modification, the surface morphology changes dramatically. The surface of PP-g-AOS becomes rough as shown in Fig. 3b. This change in surface morphology supports the occurrence of grafting reaction

3.3 XRD analysis

Figure 4 presents the XRD patterns of unmodified PP and PP-g-AOS. The difference of the samples in XRD pattern is obvious. The unmodified PP sample exhibits the characteristic diffraction peaks at 2 θ of 14.1°, 16.9° and 18.5°, which correspond to the (110), (040) and (130) lattice planes of the α monoclinic crystal of PP, respectively. Besides, there is a distinct peak at 2 θ of 16.1° which represents the (300) crystal plane of β -hexagonal crystal, suggesting that β crystal is formed in unmodified PP [27]. However, for PP-g-AOS, the (300) plane of β -hexagonal crystal disappears, moreover, the diffraction intensities of the peaks at 2 θ of 14.1°, 16.9°

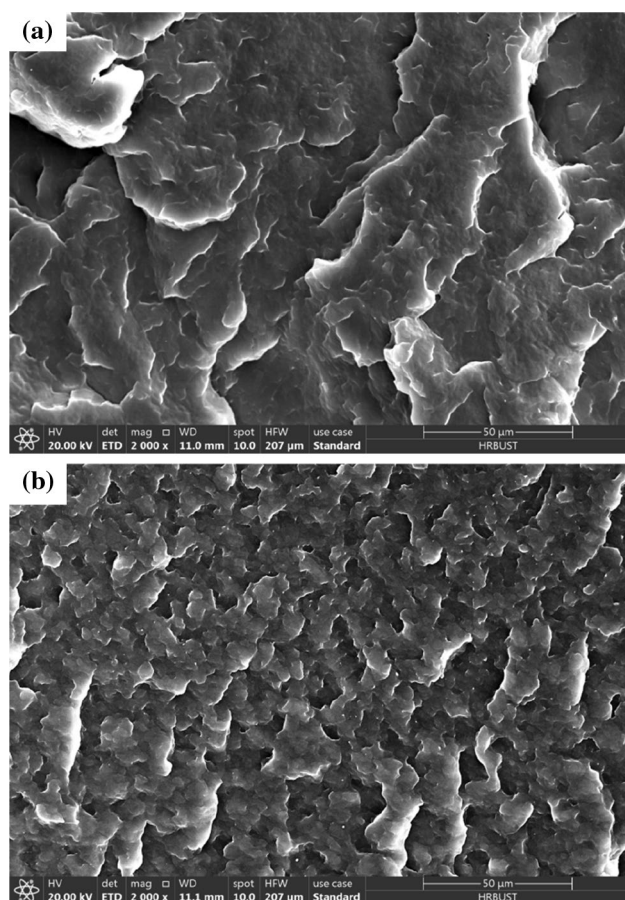


Fig. 3 SEM images for **a** unmodified PP and **b** PP-g-AOS

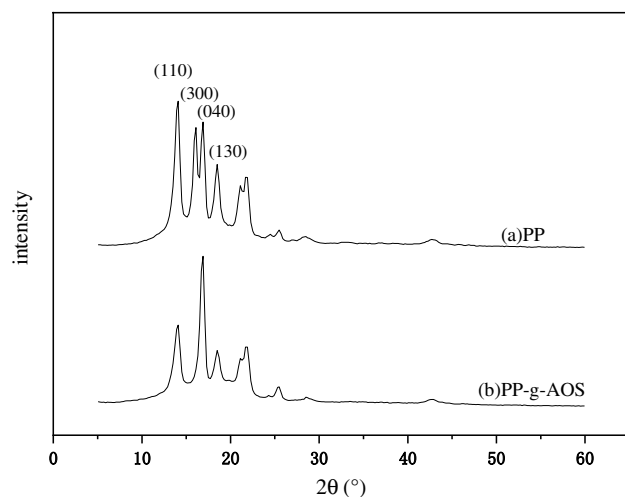


Fig. 4 XRD patterns of PP and PP-g-AOS

and 18.5° are different from those of unmodified PP. This highlights clearly that grafting alters the crystalline structure of PP, implying that AOS has been grafted onto the PP.

3.4 Effect of the temperature, screw speed, time, monomer content and initiator content on degree of grafting

To study the effect of temperature on the degree of grafting, the other reaction variables were held constant: amount of AOS, 1.4 g; amount of DCP, 0.08 g; reaction time, 10 min; screw speed, 80 r/min and the temperature was adjusted. Grafting degree of PP-g-AOS as a function of temperature is shown in Fig. 5. It can be seen that the grafting degree increases with the increase in temperature until it reaches a maximum value at 190°C and falls sharply as temperature is further increased. The experimental results can be explained as follows. With the increase of temperature, the decomposition rate of initiator is fast, and the resulting free radical concentration increases, which strengthens the ability of the offensive to macromolecules. Thus, the grafting degree increases. However, the grafting degree decreases as the temperature increases when the temperature exceeds 190°C , which is due to the increased homopolymerization of the monomers with increased temperature as well as the increased macroradical motion, resulting in increased termination of radicals [28–30]. Consequently the amount of the available monomer for grafting reaction is reduced, and the grafting degree is decreased. Therefore, the more proper temperature used should be 190°C .

The effect of the screw speed on degree of grafting was investigated while the other variables were held constant: amount of AOS, 1.4 g; amount of DCP, 0.08 g; reaction time, 10 min; temperature, 190°C and the screw speed was adjusted. Grafting degrees of PP-g-AOS as a function of screw speed is shown in Fig. 6. It can be seen from Fig. 6 that the degree of grafting increases with the increasing of screw speed. Higher screw speed could produce greater

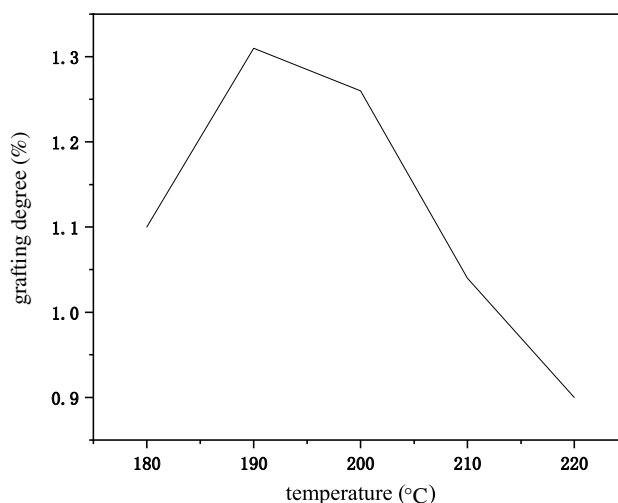


Fig. 5 Effect of temperature on grafting degree

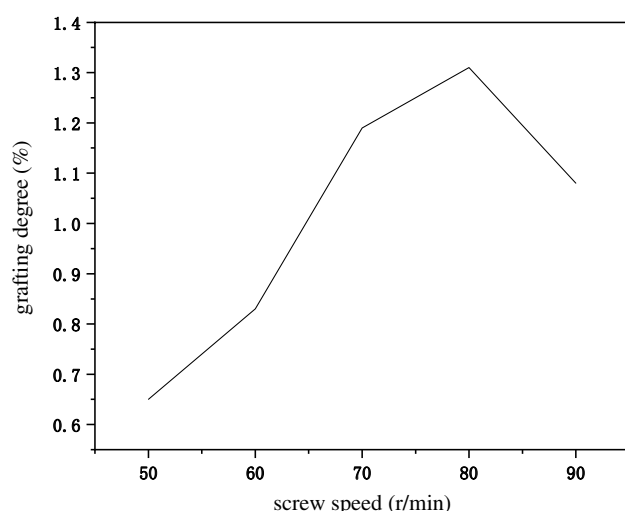


Fig. 6 Effect of screw speed on grafting degree

shear force, which is conducive to the dispersion of reactants. However, the grafting degree decreases remarkably when the screw speed exceeds 80 r/min. This can be explained that excessive shear force increased the reaction temperature as a result of increasing the shear viscous heat. This, in turn, increases homopolymerization of the monomers and also termination of radicals [31, 32]. Therefore, the more proper screw speed used should be 80 r/min.

The effect of reaction time on degree of grafting was investigated while the other variables were held constant: amount of AOS, 1.4 g; amount of DCP, 0.08 g; screw speed, 80 r/min; temperature, 190 °C and the reaction time was adjusted. Grafting degrees of PP-g-AOS as a function of reaction time is shown in Fig. 7. It was obvious that an increase in reaction time causes a significant increase in

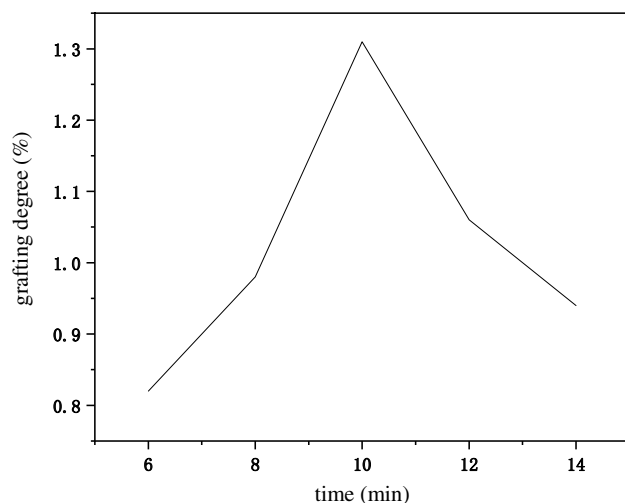


Fig. 7 Effect of reaction time on grafting degree

grafting degree in the beginning. The reason for this phenomenon is that with the increase of time, the components are mixed more uniformly and the reaction is more sufficient. But when the time is more than 10 min, the grafting degree shows a downward trend which is probably due to the side reaction of longer formed grafts [33]. Therefore, the more proper reaction time used should be 10 min.

The effect of amount of monomer on grafting degree was investigated while the other variables were held constant: amount of DCP, 0.08 g; screw speed, 80 r/min; temperature, 190 °C, reaction time, 10 min and the amount of AOS was adjusted. The relationship between grafting degree of PP-g-AOS and the amount of monomer used is given in Fig. 8. It can be seen that with the increase of amount of monomer, grafting degree increases remarkably; however, the grafting degree decreases slightly when the amount of monomer exceeds 1.4 g. The experimental result may be explained as follows. When the amount of AOS monomer is small, with the increase of the amount of monomer, free radicals have more chance to contact with monomers and the grafting reaction is accelerated obviously, thus, a higher grafting degree is achieved. As the amount of monomer is further increased, more and more primary radicals are consumed in reaction with monomer molecule, inducing undesired monomer homopolymerization, and the number of primary radicals that attack the polypropylene backbone inducing desired grafting reaction decreases accordingly [34, 35]. As a consequence, the degree of grafting falls when the amount of the monomer is beyond 1.4 g, and the more proper amount of the monomer used should be 1.4 g.

The effect of amount of DCP initiator on grafting degree was investigated while the other variables were held constant: amount of AOS, 1.4 g; screw speed, 80 r/min; temperature, 190 °C, reaction time, 10 min and the amount of

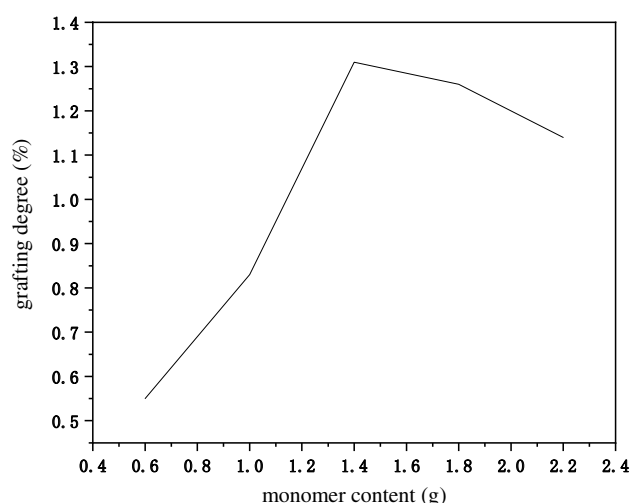


Fig. 8 Effect of monomer content on grafting degree

DCP was adjusted. The relationship between grafting degree of PP-g-AOS and the amount of DCP initiator used is given in Fig. 9. It can be seen that the degree of grafting of PP-g-AOS increases first and then decreases with increasing initiator content. When DCP content is 0.08 g, the grafting degree reaches a highest value of 1.31%. When the amount of DCP initiator is small, the increase of the initiator amount is conducive to the increase in the number of the primary radical and increase the contact chance of primary radicals and macromolecules. When the DCP content is higher than 0.08 g, the mutual termination reaction of radicals enhances, resulting in a reduction in the grafting degree [28, 34, 35]. Therefore, the more proper amount of the DCP initiator should be 0.08 g.

3.5 Space charge characteristic

Figure 8 presents the space charge distribution in PP and PP-g-AOS samples with different grafting degree during polarization under a DC electric field of 40 kV mm^{-1} applied for 1800s.

As can be seen from Fig. 10a that the pure PP used in this study shows a significant amount of hetero-charge, i.e., negative charge accumulates in the vicinity of the anode and positive charge accumulates around the cathode. The source for the hetero-charge is from ionization of impurities. Moreover, the amount of hetero-charge dramatically increases with the increase of polarization time. The results in Fig. 10b, c indicate when AOS is grafted onto PP, the hetero-charge observed decreases, besides, the accumulation of space charges reaches a stationary state in a very short time, indicating the space charge suppression occur. When the grafting degree increases to 1.14%, almost no space charge is detected inside the sample. In this case, PP-g-AOS possessed the optimal space charge suppression performance.

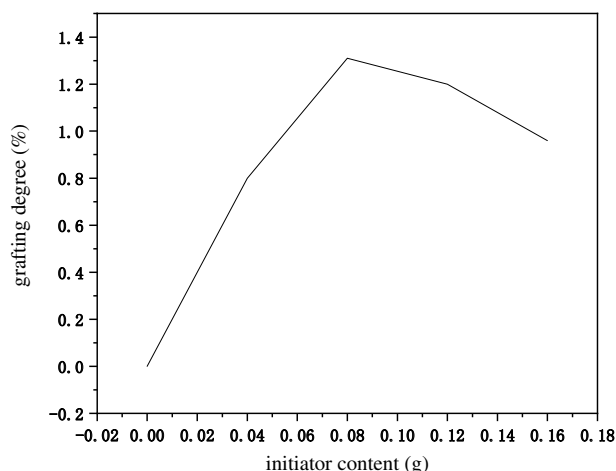


Fig. 9 Effect of DCP content on grafting degree

This is very favorable for HVDC cable insulation applications. Similar phenomena were found in LDPE-g-MAH and LDPE-g-AA [36, 37].

This feature is attributed to the enhanced trapping of injected charge. It is generally accepted that carbonyl can act as strong trapping sites of injected charge [26, 36, 37]. In consequence, as the carbonyl in AOS are introduced onto the PP chains by the grafting reaction, the homo-charge injected from the electrodes during the voltage application is trapped by the carbonyl, and neutralized the hetero-charge, resulting in decrease of hetero-charge. With the increase of grafting degree, the concentration of carbonyl increases, the corresponding charge trapping ability raise, and thus the extent of neutralizing the hetero-charge increases. The space charge is effectively suppressed. When the grafting degree of AOS exceeds 1.14% and continues to increase, more dipoles are prone to connect end to end, which reduces the number of effective charge trapping sites and cannot trap more charges. In result, the performance of space charge suppression becomes worse. As shown in Fig. 10d.

3.6 DC volume resistivity

Figure 11 illustrates the effect of grafting degree of AOS on the dc volume resistivity. With increasing grafting degree, the volume resistivity takes on a trend of increase at first, but it then slight decreases. Nevertheless, all the grafted PP samples display higher volume resistivity than that of pure PP. This feature is in accord with the results of space charge. As explained above, in the case of low grafting degree, the introduction of aromatic rings and carbonyl groups causes an increase in the number of deep traps, which can capture carriers and reduce carrier mobility, thereby increased volume resistivity is observed. As further increase of grafting degree, superfluous functional groups are linked together, which weaken the charge trapping capacity.

3.7 DC breakdown strength

Figure 12 shows the Weibull distribution of DC breakdown strength of PP and PP-g-AOS at room temperature. It can be seen that grafting AOS onto PP has a significant rise in breakdown performance. There are two possible reasons that may account for this phenomenon. Since AOS possesses similar chemical construction to the benzil-type voltage stabilizer, a similar explanation may be used here. Because of the existence of delocalized π -electrons, AOS possesses low ionizing potential and high electron affinity [38, 39], making it be capable to capture the energetic electrons by collision and to dissipate the energy, and the breakdown strength is improved. On the other hand, the energy of hot electrons is consumed through Fries rearrangement of AOS, weakening the attack of hot electrons on molecular chains, and resulting

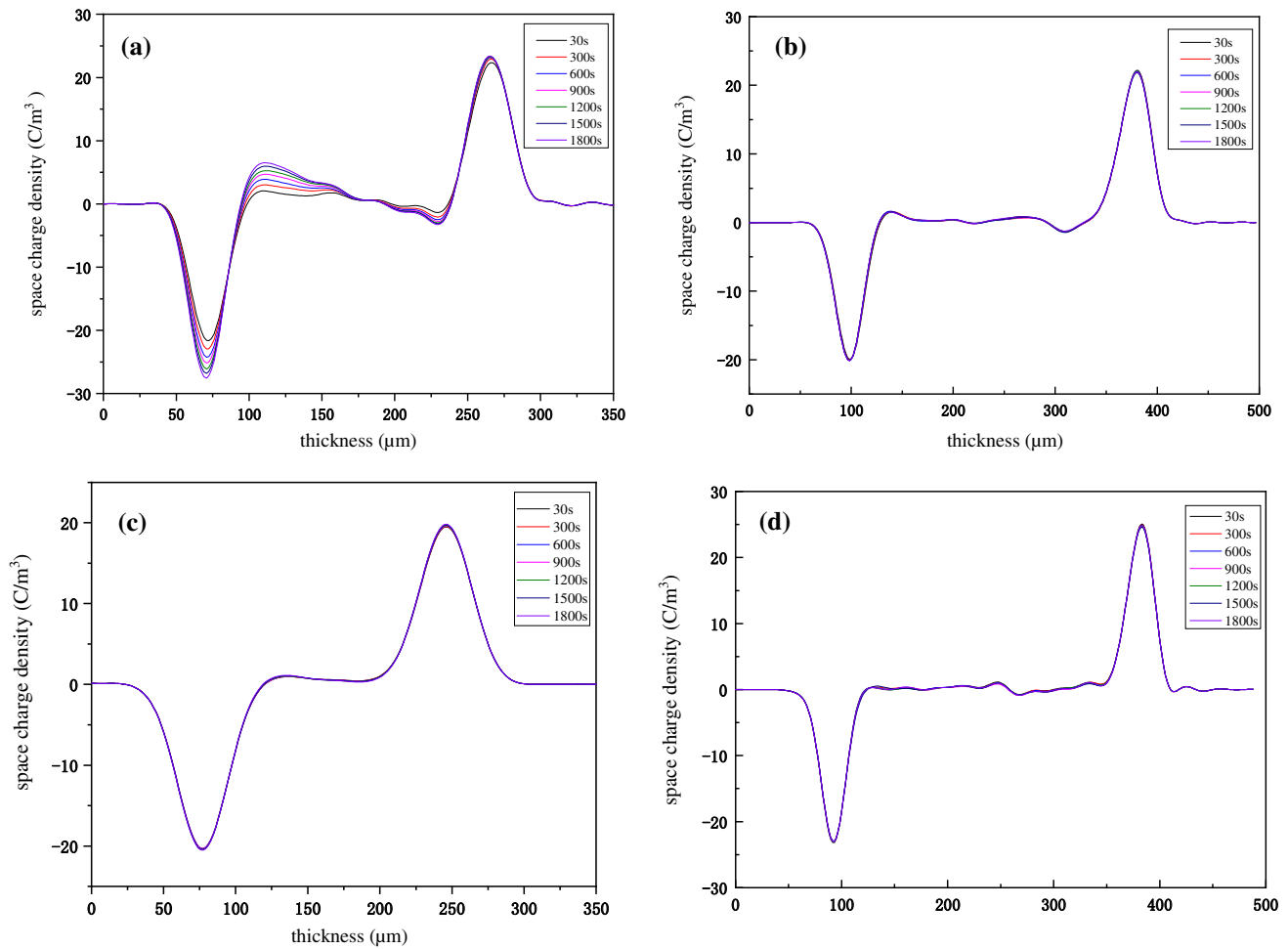


Fig. 10 Space charge distribution in PP-g-AOS samples with different grafting degree under a dc electric field of 40 kV mm^{-1} : **a** 0%, **b** 0.55%, **c** 1.14% and **d** 1.31%

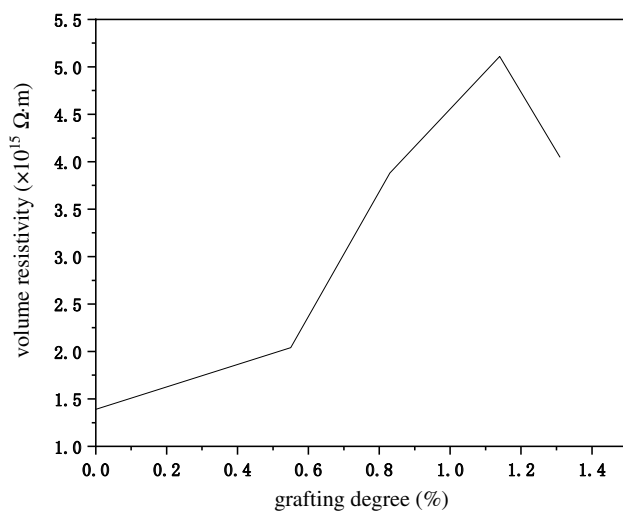


Fig. 11 Volume resistivity of PP-g-AOS with different grafting degree

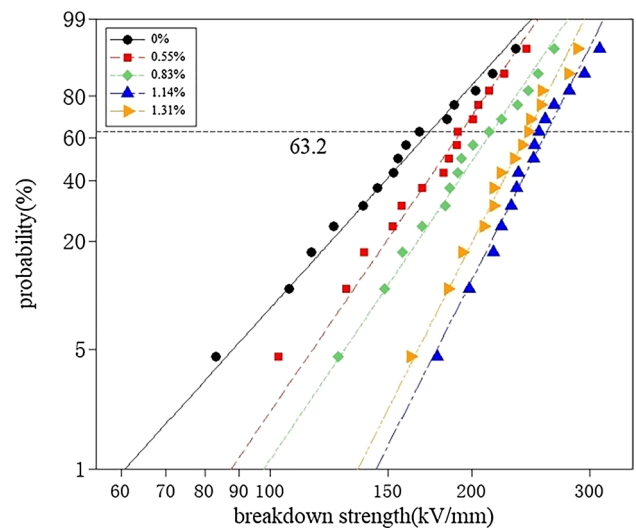


Fig. 12 Weibull distribution of the dc breakdown strength of PP-g-AOS with different grafting degree

in enhanced breakdown strength. It also can be found that when the grafting degree of PP-g-AOS is 1.31%, the breakdown strength decreases slightly, which may be related to the accumulation of space charges.

4 Conclusions

The preparation of PP-g-AOS by melt blending was described. FT-IR spectra verified that AOS was grafted onto PP molecular successfully. The degree of grafting of PP-g-AOS increased first and then decreased with increasing temperature, time, screw speed, initiator content and also the monomer content. The optimal technological parameters were determined as follows: temperature 190 °C, screw speed 80 r/min, residence time 10 min, PP:AOS:DCP = 50:1.4:0.08 (weight ratio), the maximum grafting degree of PP-g-AOS was 1.31%. Grafting AOS onto PP led to significant improvements of the space charge distribution, dc resistivity and dc breakdown strength of PP. These improvements were related to carbonyl (polar group) and benzene ring derived from AOS. It was found that PP-g-AOS with the grafting degree of 1.14% possessed the optimal electrical properties. This investigation provides a strategy for designing of polypropylene HVDC cable insulation with high electrical properties.

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References

1. T.L. Hanley, R.P. Burford, R.J. Fleming, K.W. Barber, A general review of polymeric insulation for use in HVDC cables. *IEEE Electr. Insul. Mag.* **1**, 13–24 (2003)
2. G. Mazzanti, M. Marzotto, *Extruded Cables for High Voltage Direct Current Transmission: Advances in Research and Development* (Wiley, Hoboken, 2013)
3. M.P. Bahrman, B.K. Johnson, The ABCs of HVDC transmission technologies. *IEEE Power Energy Mag.* **5**, 32–44 (2007)
4. M. Jeroense, HVDC, the next generation of transmission: highlights with focus on extruded cable systems. *IEEE Trans. Electr. Electron. Eng.* **5**, 400–404 (2010)
5. I.L. Hosier, A.S. Vaughan, S.G. Swingle, An investigation of the potential of ethylene vinyl acetate/polyethylene blends for use in recyclable high voltage cable insulation systems. *J. Mater. Sci.* **45**(9), 2747–2759 (2010). <https://doi.org/10.1158/0008-5472.CAN-03-1957>
6. G.C. Montanari, C. Laurent, G. Teyssedre, A. Campus, U.H. Nilsson, From LDPE to XLPE: investigating the change of electrical properties. Part I. space charge, conduction and lifetime. *IEEE Trans. Dielectr. Electr. Insul.* **12**, 438–446 (2005)
7. B. Dang, J.L. He, J. Hu, Y. Zhou, Tailored sPP/Silica nanocomposite for ecofriendly insulation of extruded HVDC cable. *J. Nanomater.* **2015**, 1–9 (2015)
8. I.L. Hosier, L. Cozzarini, A.S. Vaughan, S.G. Swingle, Propylene based systems for high voltage cable insulation applications. *J. Phys. Conf. Ser.* **183**, 012015 (2009)
9. I.L. Hosier, A.S. Vaughan, S.G. Swingle, An investigation of the potential of polypropylene and its blends for use in recyclable high voltage cable insulation systems. *J. Mater. Sci.* **46**, 4058–4070 (2011)
10. K. Yoshino, T. Demura, M. Kawahigashi, Y. Miyashita, K. Kurahashi, Y. Matsuda, Application of a novel polypropylene to the insulation of an electric power cable. *Electr. Eng. Jpn.* **146**, 18–26 (2004)
11. J.L. He, B. Dang, Y. Zhou, J. Hu, Reviews on research progress and key technology in extruded cables for HVDC transmission. *HighVolt. Eng.* **41**, 1417–1429 (2015)
12. J.L. He, L. PENG, Y. Zhou, Research progress of environment friendly HVDC power cable insulation materials. *HighVolt. Eng.* **43**, 337–343 (2017)
13. X.Y. Huang, Y.Y. Fan, J. Zhang, P.K. Jiang, Polypropylene based thermoplastic polymers for potential recyclable HVDC cable insulation applications. *IEEE Trans. Dielectr. Electr. Insul.* **24**, 1446–1456 (2017)
14. T.T.N. Vu, G. Teyssedre, B. Vissouvanadin, S. Le Roy, C. Laurent, Correlating conductivity and space charge measurements in multi-dielectrics under various electrical and thermal stresses. *IEEE Trans. Dielectr. Electr. Insul.* **22**, 117–127 (2015)
15. D. Fabiani, G.C. Montanari, A. Cavallini, G. Mazzanti, Relation between space charge accumulation and partial discharge activity in enameled wires under PWM-like voltage waveforms. *IEEE Trans. Dielectr. Electr. Insul.* **11**, 393–405 (2004)
16. G.C. Montanari, Bringing an insulation to failure: the role of space charge”. *IEEE Trans. Dielectr. Electr. Insul.* **18**, 339–364 (2011)
17. D. Pitsa, M.G. Danikas, G.E. Vardakis, T. Tanaka, Influence of homocharges and nanoparticles in electrical tree propagation under DC voltage application. *Electr. Eng.* **94**, 81–88 (2012)
18. Y.H. Wu, J.W. Zha, W.K. Li, S.J. Wang, Z.M. Dang, A remarkable suppression on space charge in isotactic polypropylene by inducing the β -crystal formation. *Appl. Phys. Lett.* **107**, 112901 (2015)
19. B.X. Du, H. Xu, J. Li, Z. Li, Space charge behaviors of PP/POE/ZnO nanocomposites for HVDC cables. *IEEE Trans. Dielectr. Electr. Insul.* **23**, 3165–3174 (2016)
20. Y. Zhou, J. Hu, B. Dang, J.L. He, Titanium oxide nanoparticle increases shallow traps to suppress space charge accumulation in polypropylene dielectrics. *RSC Adv.* **6**, 48720–48727 (2016)
21. G.C. Montanari, G. Teyssedre, C. Laurent, D. Kaempfer, R. Mülhaupt, Investigating charge trapping behaviour of nanocomposite isotactic and syndiotactic polypropylene matrix. *IEEE ICSD*, pp. 836–839 (2004)
22. Y. Zhou, B. Dang, J. Hu, X. Chen, F. Yu, J.L. He, Effect of magnesium oxide nanoparticles on tailoring the properties of polypropylene. *Proc. CSEE* **36**, 6619–6626 (2016)
23. W.K. Cao, Z. Li, J. Gong, G.H. Sheng, X.C. Jiang, Influence of nano-particle concentration and surface treatment on space charge behavior in nano-MgO/PP. *HighVolt. Eng.* **41**, 1495–1504 (2015)
24. H.M. Cui, H. Liu, J.Y. Wang, X. Li, F. Han, Agglomeration and dispersion of nano-scale powders. *Mater. Mech. Eng.* **28**, 38–41 (2004)
25. S.Y. Chen, G.D. Meng, B. Kong, B. Xiao, Z.D. Wang, Z. Jing, Y.S. Gao, G.L. Wu, H. Wang, Y.H. Cheng, Asymmetric alicyclic amine-polyether amine molecular chain structure for improved energy storage density of high-temperature crosslinked polymer capacitor. *Chem. Eng. J.* **387**, 123662 (2019)
26. J.W. Zha, Y.H. Wu, S.J. Wang, D.H. Wu, H.D. Yan, Z.M. Dang, Improvement of space charge suppression of polypropylene for potential application in HVDC cables. *IEEE Trans. Dielectr. Electr. Insul.* **23**, 2337–2343 (2016)

27. L.F. Cui, P.L. Wang, Y.G. Zhang, X. Zhou, L. Xu, L. Zhang, L.T. Zhang, L. Liu, X.F. Guo, Glass fiber reinforced and β -nucleating agents regulated polypropylene: a complementary approach and a case study. *J. Appl. Polym. Sci.* **135**, 45768 (2017)
28. N. Işıklan, F. Kurşun, M. İnal, Graft copolymerization of itaconic acid onto sodium alginate using benzoyl peroxide. *Carbohydr. Polym.* **79**, 665–672 (2010)
29. T. Sun, P.X. Xu, Q. Liu, J. Xue, W.M. Xie, Graft copolymerization of methacrylic acid onto carboxymethyl chitosan. *Eur. Polym. J.* **39**, 189–192 (2003)
30. Y. Pan, J. Ruan, D. Zhou, Solid-phase grafting of glycidyl methacrylate onto polypropylene. *J. Appl. Polym. Sci.* **65**, 1905–1912 (1997)
31. M.K. Razavi Aghjeh, H. Nazockdast, H. Assempour, Parameters affecting the free-radical melt grafting of maleic anhydride onto linear low-density polyethylene in an internal mixer. *J. Appl. Polym. Sci.* **99**, 141–149 (2005)
32. H. Cartier, G.H. Hu, Plastification or melting: a critical process for free radical grafting in screw extruders. *Polym. Eng. Sci.* **38**, 177–185 (1998)
33. T.H. Kim, D.R. Oh, Melt grafting of maleimides having hindered phenol antioxidant onto low molecular weight polyethylene. *Polym. Degrad. Stab.* **84**, 499–503 (2004)
34. D.A. Shi, J.H. Yang, Z.H. Yao, Y. Wang, H.L. Huang, W. Jing, J.H. Yin, G. Costa, Functionalization of isotactic polypropylene with maleic anhydride by reactive extrusion: mechanism of melt grafting. *Polymer* **42**, 5549–5557 (2001)
35. A. Zare, M. Morshed, R. Bagheri, K. Karimi, Effect of various parameters on the chemical grafting of amide monomers to poly(lactic acid). *Fibers Polym.* **14**, 1783–1793 (2013)
36. K.S. Suh, C.R. Lee, Y. Zhu, Electrical properties of chemically modified polyethylenes. *IEEE Trans. DEI.* **4**, 681–687 (1997)
37. S.H. Lee, J.K. Park, J.H. Han, K.S. Suh, Space charge and electrical conduction in maleic anhydride-grafted polyethylene. *IEEE Trans. Dielectr. Electron.* **2**, 1132–1139 (1995)
38. H. Zhang, Y. Shang, H. Zhao, B. Han, Z. Li, Study of the effect of valence bond isomerizations on electrical breakdown by adding acetophenone to polyethylene as voltage stabilizers. *Comput. Theor. Chem.* **1062**, 99–104 (2015)
39. H. Zhang, Y. Shang, H. Zhao, X. Wang, B. Han, Z. Li, Theoretical study on the tailored side-chain architecture of benzil-like voltage stabilizers for enhanced dielectric strength of cross-linked polyethylene. *RSC Adv.* **6**, 11618–11630 (2016)

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